

Scalability & Future Plan

Date	02 July 2026
Team ID	
Project Name	Rising Waters – A Machine Learning Approach to Flood Prediction
Maximum Marks	1 Mark

Scalability & Future Plan

This document outlines how the current Rising Waters solution can be scaled to handle larger user bases, increased data loads, or extended features in the future, and captures the team's roadmap for enhancing the project beyond its current state.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Small training dataset (115 labelled flood records)	Model may not generalize well to extreme or unseen weather patterns	High
2	Manual data entry for predictions (no live weather feed)	Slower workflow and possible data-entry error during time-critical events	Medium
3	Single-region focus (India, 36 subdivisions)	Not directly usable for other countries without retraining	Medium
4	No user authentication / prediction history	Cannot track usage per coordinator or audit past predictions	Low

Scalability Plan

Scalability Aspect	Current State	Proposed Upgrade / Solution
User Load	Single Unicorn instance	Horizontal scaling via IBM Cloud Code Engine auto-scaling or multiple Cloud Foundry instances
Data Storage	Static local Excel files + joblib bundle	Move source data and model bundle to IBM Cloud Object Storage for versioned, shared access
Prediction Throughput	Single-model inference per request	Add response caching for repeated identical inputs; explore batch-prediction API for bulk analysis
Data Freshness	Manual retraining via notebook re-run	Automate retraining via a scheduled pipeline (e.g., IBM Cloud Continuous Delivery) when new labelled data arrives

Future Roadmap

- Integrate a live weather/rainfall API so fields can be auto-populated instead of manually entered.
- Expand the labelled flood dataset beyond 115 records (more years / more subdivisions) to improve model robustness and reduce variance from small-sample train/test splits.
- Add subdivision-level historical comparison charts directly into the web app (reusing the national rainfall EDA already built in the notebook).
- Introduce basic user accounts so coordinators can save and review past predictions over a season.
- Extend the model to a regression output (expected rainfall volume) in addition to the current binary flood/no-flood classification, for finer-grained risk assessment.